Computer Architecture (CSC213)

B.Sc. CSIT — 3rd Semester | Tribhuvan University

Important Questions + Study Focus + Key Concepts | Past Papers 2075–2081

■ **Very High** — 4+ times	■ **High** — 3 times	■ **Medium** — 2 times	■ **Low** — 1 time / Syllabus
★ = Must Prepare (3+ appearances)	**[Long]** = 10 marks	**[Short]** = 5 marks	■ = Focus Topic | ✏■ = Study Task

■ Unit Priority Overview

Unit	Topic	Priority	Key Exam Focus
U1	Data Representation	■ Medium	Parity circuit, signed numbers, overflow detection
U2	Register Transfer	■ Medium	RTL, adder-subtractor, shift microops
U3	Basic Computer	■ Very High	Common bus, instruction format, FETCH RTL, data transfer
U4	Microprog. Control	■ Very High	Hardwired vs microprog, sequencer diagram
U5	CPU	■ High	Addressing modes, RISC/CISC, address instruction programs
U6	Pipelining	■ Very High	4-seg pipeline, hazards, Flynn's classification, speedup
U7	Computer Arithmetic	■ Very High	Booth multiplication, restoring/non-restoring division
U8	I/O Organization	■ High	DMA, priority interrupt, programmed I/O comparison
U9	Memory Organization	■ High	Cache mapping, hit ratio, direct vs associative

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UNIT 1 | Data Representation — 4 Hours

■ Past Exam Questions

• **Q1.** How is parity bit generated in even parity? Demonstrate with table and circuit diagram.	**[Short]** ■ LOW Years: 2081
◆ **Q2.** Explain error detection codes with example (Parity bit, odd/even parity, generator & checker).	**[Short]** ■ MEDIUM Years: 2075-1, 2078
• **Q3.** What do you mean by error correction codes? Explain any one technique.	**[Short]** ■ LOW Years: 2080
• **Q4.** What are different methods for representing signed numbers? Represent (-71) in those formats.	**[Short]** ■ LOW Years: 2079
• **Q5.** Differentiate between floating point and fixed point representation.	**[Short]** ■ LOW Years: 2079
◆ **Q6.** What is arithmetic overflow? How can it be detected?	**[Short]** ■ MEDIUM Years: 2080, 2081

■ FOCUS TOPICS — What to concentrate on while studying

- ■ Parity Bit (Odd & Even) — generator and checker circuit
- ■ Signed Number Representation — Sign-Magnitude, 1's Complement, 2's Complement
- ■ Floating Point vs Fixed Point Representation

■ Overflow detection in arithmetic operations

■ Gray Code, Excess-3 Code, BCD — differences and conversions

■ WHAT TO STUDY — Key things to prepare for this unit

═■ Draw parity generator/checker circuit for 4-bit input

═■ Convert any decimal to Sign-Magnitude, 1's complement, 2's complement

═■ IEEE 754 floating point format (sign, exponent, mantissa)

═■ Know overflow detection rule: carry-in ≠ carry-out at MSB

═■ Practice BCD ↔ Binary ↔ Gray Code conversions

UNIT 2 | Register Transfer and Microoperations — 5 Hours

■ Past Exam Questions

◆ **Q7.** What do you mean by Register Transfer Language? Explain RTL control function with example.	**[Short]** ■ MEDIUM Years: 2079, 2077
★ **Q8.** What is micro-operation? Explain different arithmetic microoperations.	**[Short]** ■ HIGH Years: 2080, 2075-1, 2081
◆ **Q9.** Explain the binary adder-subtractor circuit with suitable diagram.	**[Short]** ■ MEDIUM Years: 2079, 2078
◆ **Q10.** Explain different types of shift microoperations (logical, circular, arithmetic).	**[Short]** ■ MEDIUM Years: 2081, 2075-1
◆ **Q11.** Explain logic microoperations and their hardware implementation.	**[Short]** ■ MEDIUM Years: 2075-1, 2080

■ FOCUS TOPICS — What to concentrate on while studying

■ Register Transfer Language (RTL) notation and control functions

■ Arithmetic Microoperations — Binary Adder, Adder-Subtractor, Incrementer

■ Logic Microoperations — AND, OR, XOR, Complement with 16 operations table

■ Shift Microoperations — Logical Shift, Circular Shift, Arithmetic Shift

■ Hardware implementation of shifter (combinational circuit)

■ WHAT TO STUDY — Key things to prepare for this unit

═■ Write RTL expressions for register transfers with control signals

═■ Draw 4-bit binary adder-subtractor using XOR + full adders

═■ Know all 4 types of shift — left/right for each type

═■ Memorize 16 logic microoperations table (AND, OR, XOR, NOT etc.)

═■ Arithmetic circuit: draw and explain selection of operation

UNIT 3 | Basic Computer Organization and Design — 8 Hours

■ Past Exam Questions

★ **Q12.** Explain common bus system of basic computer with a diagram.	**[Short]** ■ VERY HIGH Years: 2081, 2079, 2080, 2077
★ **Q13.** What is instruction format? Explain instruction formats of basic computer with examples.	**[Short]** ■ VERY HIGH Years: 2081, 2080, 2078, 2075-1

◆ **Q14.** Explain the instruction format of basic computer. Write symbolic representation for FETCH routine.	[Long] ■ MEDIUM Years: 2080, 2081
◆ **Q15.** What microoperations are performed in fetch phase of instruction cycle? Explain with diagram.	[Short] ■ MEDIUM Years: 2081, 2077
◆ **Q16.** Draw instruction cycle state diagram with interrupt and explain it.	[Short] ■ MEDIUM Years: 2077, 2078
★ **Q17.** Explain the data transfer instructions with example.	[Short] ■ VERY HIGH Years: 2078, 2075-1, 2077, 2079
◆ **Q18.** Explain stored-program concept with example.	[Short] ■ MEDIUM Years: 2075, 2075-1
◆ **Q19.** Explain program interrupt and interrupt cycle of basic computer.	[Short] ■ MEDIUM Years: 2075-1, 2081

■ FOCUS TOPICS — What to concentrate on while studying

- ■ Common Bus System — 16 registers, bus lines, selection signals (S0,S1,S2)
- ■ Instruction Format — Memory Reference, Register Reference, I/O Instructions
- ■ Instruction Cycle — Fetch, Decode, Execute phases with RTL
- ■ FETCH routine: T0: AR←PC, T1: IR←M[AR], PC←PC+1, T2: Decode
- ■ Program Interrupt — interrupt cycle, IEN flag, R flag
- ■ Memory Reference Instructions — ADD, AND, LDA, STA, BUN, BSA, ISZ
- ■ Input-Output Instructions — INP, OUT, SKI, SKO, ION, IOF

■ WHAT TO STUDY — Key things to prepare for this unit

- ⇒■ Draw complete common bus system with all 16 registers and MUX
- ⇒■ Write ALL memory reference instruction operations in RTL
- ⇒■ Memorize FETCH cycle RTL: T0→T1→T2 exactly
- ⇒■ Draw instruction cycle flowchart including interrupt
- ⇒■ Know instruction format: bit positions (0-11 address, 12 I-bit, 13-15 opcode)
- ⇒■ Practice decoding: how the computer determines instruction type

UNIT 4 | Microprogrammed Control — 4 Hours

■ Past Exam Questions

★ **Q20.** Differentiate between hardwired and microprogrammed control unit.	[Short] ■ VERY HIGH Years: 2078, 2077, 2075, 2075-1, 2079, 2080
★ **Q21.** Describe with example the microprogram sequencer used in microprogrammed control.	[Long] ■ HIGH Years: 2078, 2079, 2080
★ **Q22.** Explain block diagram of microprogram sequencer.	[Short] ■ HIGH Years: 2080, 2079, 2078
◆ **Q23.** Explain the symbolic microinstruction with example.	[Short] ■ MEDIUM Years: 2078, 2081
◆ **Q24.** What do you mean by sequencer? Explain with microprogram sequencer.	[Short] ■ MEDIUM Years: 2079, 2081
• **Q25.** Explain address sequencing and conditional branch in microprogrammed control.	[Short] ■ LOW Years: 2079

■ **FOCUS TOPICS — What to concentrate on while studying**

- ■ Hardwired Control — combinational logic, fast but inflexible
- ■ Microprogrammed Control — control word, control memory, microprogram
- ■ Microprogram Sequencer — CAR (Control Address Register), SBR, MUX, incrementer
- ■ Address Sequencing — increment, conditional branch, mapping, subroutine
- ■ Microinstruction Format — fields: F1, F2, F3, CD, BR, AD
- ■ Symbolic Microinstruction — labels, operations, condition, branch address
- ■ Control Word — each bit controls a microoperation

■ **WHAT TO STUDY — Key things to prepare for this unit**

- ═■ Draw microprogram sequencer block diagram (must memorize)
- ═■ Comparison table: Hardwired vs Microprogrammed (speed, flexibility, cost, design)
- ═■ Know microinstruction format fields and their meaning
- ═■ Write symbolic microinstructions for FETCH routine
- ═■ Understand 4 address sequencing methods
- ═■ Know how instruction opcode maps to control memory address

UNIT 5 | Central Processing Unit — 4 Hours

■ Past Exam Questions

★ **Q26.** Explain various types of addressing modes with example.	**[Short]** ■ **VERY HIGH** Years: 2079, 2078, 2077, 2075, 2081
★ **Q27.** Differentiate between RISC and CISC architecture.	**[Short]** ■ **VERY HIGH** Years: 2080, 2079, 2075, 2075-1, 2081
◆ **Q28.** In RISC architecture, explain overlapping register windows.	**[Long]** ■ **MEDIUM** Years: 2075, 2079
★ **Q29.** Write program using 3-address, 1-address, and 0-address instructions for: X=(A+B*C-D)/(E*F+G)	**[Short]** ■ **HIGH** Years: 2080, 2079, 2077
◆ **Q30.** Write short notes on: a) CISC b) Conditional branch	**[Short]** ■ **MEDIUM** Years: 2081, 2075-1
◆ **Q31.** Explain types of interrupts.	**[Short]** ■ **MEDIUM** Years: 2078, 2081

■ **FOCUS TOPICS — What to concentrate on while studying**

- ■ Addressing Modes — Immediate, Direct, Indirect, Register, Register Indirect, Displacement, Relative, Indexed, Base-Register
- ■ RISC vs CISC — comparison on all parameters
- ■ Overlapped Register Windows in RISC (SPARC architecture)
- ■ Instruction Formats — 3-address, 2-address, 1-address, 0-address (stack)
- ■ Subroutine Call and Return mechanism
- ■ Types of Interrupt — external, internal, software (trap)

■ **WHAT TO STUDY — Key things to prepare for this unit**

- ═■ Know each addressing mode with EA (Effective Address) formula and example

═■ Draw overlapping register windows diagram for RISC

═■ RISC vs CISC table: instructions, registers, clock cycles, addressing modes, code size

═■ Practice writing same expression in 3-addr, 2-addr, 1-addr, 0-addr form

═■ Know advantages: RISC — pipeline friendly, CISC — compact code

UNIT 6 | Pipelining

6 Hours

■ Past Exam Questions

★ **Q32.** What is pipelining? Explain 4-segment instruction pipeline. What are its advantages?	[Long] ■ HIGH Years: 2081, 2080, 2077
◆ **Q33.** Explain with space-time diagram for 6-segment pipeline processing 8 tasks.	[Long] ■ MEDIUM Years: 2077, 2075
★ **Q34.** What are different types of pipeline hazards? Explain each with example.	[Long] ■ VERY HIGH Years: 2080, 2079, 2075, 2081
★ **Q35.** What do you understand by Flynn's classification? Explain.	[Short] ■ VERY HIGH Years: 2081, 2080, 2075, 2078
◆ **Q36.** How is performance of computer increased using pipeline? Explain speedup equation.	[Short] ■ MEDIUM Years: 2077, 2081
◆ **Q37.** Differentiate between instruction pipeline and arithmetic pipeline.	[Short] ■ MEDIUM Years: 2075, 2077
• **Q38.** Explain vector processing and vector operations.	[Short] ■ LOW Years: 2078

■ FOCUS TOPICS — What to concentrate on while studying

■ Pipeline concept — stages, throughput, speedup formula: Speedup = n×k / (k+n-1)

■ 4-Segment Instruction Pipeline: FI→DI→CO→EI (Fetch, Decode, Calc Operand, Execute)

■ Space-Time Diagram — draw for k segments, n tasks

■ Pipeline Hazards: Structural, Data (RAW/WAR/WAW), Control hazards + solutions

■ Flynn's Classification: SISD, SIMD, MISD, MIMD with diagrams

■ Instruction vs Arithmetic Pipeline — differences and examples

■ Floating point pipelining — addition/subtraction stages

■ WHAT TO STUDY — Key things to prepare for this unit

═■ Draw space-time diagram for 4-segment, 6 tasks — calculate throughput

═■ Memorize speedup formula and solve numericals

═■ Draw all 4 Flynn's classification diagrams (SISD, SIMD, MISD, MIMD)

═■ Know 3 hazard types with specific examples and solutions (forwarding, stall, branch prediction)

═■ Draw 4-segment instruction pipeline stages clearly

═■ Floating point addition pipeline: 6 stages (compare, shift, add, normalize, round, output)

UNIT 7 | Computer Arithmetic

6 Hours

■ Past Exam Questions

★ **Q39.** Explain Booth's multiplication algorithm. Perform 50×(-13) using Booth algorithm.	**[Long]** ■ **VERY HIGH** Years: 2080, 2077, 2075-1, 2081
◆ **Q40.** Multiply (-4)×(-3) using Booth multiplication algorithm with hardware diagram.	**[Long]** ■ **MEDIUM** Years: 2077, 2078
◆ **Q41.** Draw flowchart for addition/subtraction of signed 2's complement. Perform (90-43).	**[Long]** ■ **MEDIUM** Years: 2081, 2079
★ **Q42.** Explain non-restoring division with flowchart and hardware. Divide 10/3 using restoring division.	**[Long]** ■ **HIGH** Years: 2078, 2075-1, 2079
◆ **Q43.** Differentiate between restoring and non-restoring division.	**[Short]** ■ **MEDIUM** Years: 2077, 2078
◆ **Q44.** Divide 10/4 using non-restoring division.	**[Short]** ■ **MEDIUM** Years: 2075, 2079
• **Q45.** Addition and subtraction with signed magnitude data.	**[Short]** ■ **LOW** Years: 2081

■ FOCUS TOPICS — What to concentrate on while studying

- ■ Signed 2's Complement Addition/Subtraction — flowchart and overflow detection
- ■ Booth Multiplication Algorithm — Qn+1 bit, Q0Q(n+1) cases: 00,01,10,11
- ■ Booth Hardware — AC, QR, Q(n+1), SC registers + ALU
- ■ Restoring Division Algorithm — restore partial remainder if negative
- ■ Non-restoring Division — add or subtract based on sign, no restoration
- ■ Divide Overflow detection
- ■ Signed Magnitude multiplication — partial products

■ WHAT TO STUDY — Key things to prepare for this unit

- ═■ Memorize Booth algorithm: 4 cases based on Qn and Qn+1 bits
- ═■ Practice Booth with at least 3 examples (both negative, both positive, mixed)
- ═■ Draw Booth hardware implementation diagram
- ═■ Practice restoring division: step-by-step with partial remainders
- ═■ Practice non-restoring division: know when to add vs subtract
- ═■ Draw flowcharts for both division algorithms
- ═■ Signed 2's complement addition flowchart — 3 decision points

UNIT 8 | Input Output Organization

4 Hours

■ Past Exam Questions

★ **Q46.** What do you understand by priority interrupt? Explain polling, daisy-chaining and parallel priority interrupt.	**[Long]** ■ **HIGH** Years: 2081, 2078, 2077
★ **Q47.** What is DMA? Explain DMA controller with block diagram. How does DMA interact with I/O device?	**[Long]** ■ **VERY HIGH** Years: 2080, 2077, 2075, 2079
★ **Q48.** Define I/O interface. Compare programmed I/O, interrupt-driven I/O, and DMA.	**[Long]** ■ **HIGH** Years: 2077, 2075-1, 2081
◆ **Q49.** Why is I/O interface important? Discuss programmed I/O with flowchart.	**[Short]** ■ **MEDIUM** Years: 2081, 2075-1

★ **Q50.** Differentiate between IOP and DMA.	[Short] ■ HIGH Years: 2078, 2075-1, 2079
◆ **Q51.** Differentiate between isolated vs memory-mapped I/O.	[Short] ■ MEDIUM Years: 2077, 2079
◆ **Q52.** Explain asynchronous data transfer: Strobe and Handshaking.	[Short] ■ MEDIUM Years: 2078, 2081

■ FOCUS TOPICS — What to concentrate on while studying

- ■ Priority Interrupt — Polling (software), Daisy-Chaining (hardware), Parallel Priority (hardware with encoder)
- ■ DMA Controller — block diagram, DMA cycle steal, burst mode, transparent mode
- ■ Programmed I/O — CPU polls status flag (busy wait)
- ■ Interrupt-Initiated I/O — CPU interrupted when device ready
- ■ I/O Processor (IOP) — independent processor for I/O, has own memory access
- ■ Memory-Mapped I/O vs Isolated I/O — same vs separate address space
- ■ Strobe vs Handshaking — one-way vs two-way synchronization

■ WHAT TO STUDY — Key things to prepare for this unit

- ═■ Draw DMA controller block diagram with all internal registers
- ═■ Draw daisy-chaining diagram and explain BG/BR signals
- ═■ Comparison table: Programmed I/O vs Interrupt I/O vs DMA (efficiency, CPU involvement)
- ═■ Draw priority interrupt with daisy chain — know how ISR address is found
- ═■ Know difference: IOP vs DMA — IOP is like a separate CPU, DMA just transfers data
- ═■ Flowchart for programmed I/O — polling loop

UNIT 9 | Memory Organization

4 Hours

■ Past Exam Questions

★ **Q53.** What is cache memory? Explain mapping process. Differentiate direct mapping and associative mapping.	[Long] ■ HIGH Years: 2079, 2075-1, 2081
◆ **Q54.** What is meant by cache mapping? Explain direct mapping with block diagram.	[Short] ■ MEDIUM Years: 2081, 2078
★ **Q55.** Explain cache memory: locality of reference, hit & miss ratio, mapping, write policies.	[Short] ■ HIGH Years: 2080, 2081, 2079
★ **Q56.** What are advantages and disadvantages of direct mapping vs associative mapping?	[Short] ■ HIGH Years: 2078, 2075, 2079
◆ **Q57.** Why is replacement algorithm required in associative mapping?	[Short] ■ MEDIUM Years: 2077, 2078
◆ **Q58.** Define associative memory. Explain with block diagram.	[Short] ■ MEDIUM Years: 2075, 2078
◆ **Q59.** What are main characteristics of memory system? Explain memory hierarchy.	[Short] ■ MEDIUM Years: 2075-1, 2079

■ FOCUS TOPICS — What to concentrate on while studying

- ■ Memory Hierarchy — registers → cache → RAM → disk (speed vs cost vs size)
- ■ Cache Memory — why needed, locality of reference (temporal & spatial)

- ■ Hit Ratio & Miss Ratio — formulas: h = hits/total access; $T_{avg} = h \times T_c + (1-h) \times T_m$
- ■ Direct Mapping — fixed cache line for each memory block (line = block mod cache_size)
- ■ Associative Mapping — any block → any line; needs replacement algorithm (LRU, FIFO, Random)
- ■ Set-Associative Mapping — compromise between direct and associative
- ■ Write Policies — Write-Through vs Write-Back
- ■ Associative Memory — content-addressable memory, match logic

■ WHAT TO STUDY — Key things to prepare for this unit

- ⇒■ Draw memory hierarchy pyramid with access times
- ⇒■ Solve hit ratio problems: given h and cache/main memory time → find avg access time
- ⇒■ Draw direct mapping: show address division (tag | line | word)
- ⇒■ Draw associative mapping: CAM-based lookup, replacement algorithm
- ⇒■ Comparison table: Direct vs Associative vs Set-Associative mapping
- ⇒■ Know LRU replacement algorithm with example
- ⇒■ Write-through vs Write-back: know pros/cons of each

■■ Complete Exam Strategy

■ MUST-PREPARE LIST (appeared 3+ times — very likely to come)

→ Pipelining — 4-segment instruction pipeline, space-time diagram, speedup formula

→ Booth Multiplication — algorithm, hardware diagram, numerical problems

→ Common Bus System — diagram with all 16 registers

→ Hardwired vs Microprogrammed Control Unit — comparison table

→ Addressing Modes — all 8-9 modes with EA formula and example

→ Cache Memory Mapping — direct vs associative vs set-associative

→ DMA Controller — block diagram, interaction with CPU and I/O

→ Priority Interrupt — polling, daisy-chaining, parallel priority with diagrams

→ Instruction Format of Basic Computer — 3 types with examples

→ Non-restoring Division & Booth Multiplication — practice 3+ numericals

■ SECTION A STRATEGY (Long questions — attempt any 2 of 3 — 10 marks each)

→ Always draw diagrams — bare text gets 40-50% marks at most

→ Unit 6 (Pipelining) — almost always has one long question; prepare pipeline diagrams

→ Unit 7 (Computer Arithmetic) — Booth multiplication or division is almost certain

→ Unit 3/4 (Basic Computer / Microprog) — common bus system or sequencer

→ Unit 8 (I/O) — DMA or Priority Interrupt as long question

→ Unit 9 (Memory) — Cache mapping comes frequently as long question

→ Always show step-by-step working for numerical problems (Booth, division)

■ SECTION B STRATEGY (Short questions — attempt any 8 of 9 — 5 marks each)

→ Pick the 8 you know best — do NOT attempt all 9

→ Unit 2 (RTL, shift ops, adder) — easy 5 marks if diagram is clear

→ Unit 5 (RISC vs CISC, addressing modes) — very predictable short questions

→ Unit 1 (Parity, signed numbers) — straightforward if practiced

→ Flynn's Classification — draw all 4 diagrams = easy 5 marks

→ Short notes questions are common: CISC, Conditional Branch, Flynn's, Common Bus

→ Address instruction programs (3-addr / 1-addr / 0-addr) — practice expression X=...

■ NUMERICAL PROBLEMS — PRACTICE THESE

→ Booth Multiplication: 50×(-13), (-4)×(-3), any 4-bit × 4-bit combination

→ Non-restoring Division: 10÷3, 10÷4, 5÷3 (step-by-step)

→ Restoring Division: 23÷9 (from 2079 paper)

→ Signed 2's complement: 90-43, addition and subtraction with overflow check

→ Pipeline speedup: given k segments and n tasks → find time and speedup

→ Hit ratio: Tavg = h×Tc + (1-h)×Tm — solve for different h values

→ Address instruction programs: X = (A+B*C-D)/(E*F+G) in 3-addr, 1-addr, 0-addr

■ DIAGRAMS TO MEMORIZE (draw these without looking)

→ Common Bus System of Basic Computer (16 registers + bus lines)

→ Microprogram Sequencer block diagram

→ 4-segment Instruction Pipeline (FI, DI, CO, EI stages)

→ Space-Time Diagram for pipeline (draw as grid)

→ Flynn's Classification: SISD, SIMD, MISD, MIMD

→ DMA Controller block diagram

→ Daisy-chain priority interrupt circuit

→ Direct Mapping and Associative Mapping block diagrams

→ Binary Adder-Subtractor circuit (4-bit)

→ Booth Multiplication hardware (AC, QR, Qn+1, SC)
